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User & working directory

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Practical exercise submission covering course pages 15-17. The screenshots document commands run in a Kali Linux virtual machine; the answers below summarize those captured results.



Figure 1. User identity, hostname and current directory.

whoami: The logged-in username is `kali`.

hostname: The machine hostname is `kali`.

pwd: The current working directory is `/home/kali`.

These commands identify the active user, the machine name and the current location in the filesystem.

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Operating system & CPU

`uname -a` displays kernel and system information. `lscpu` displays processor and architecture details.

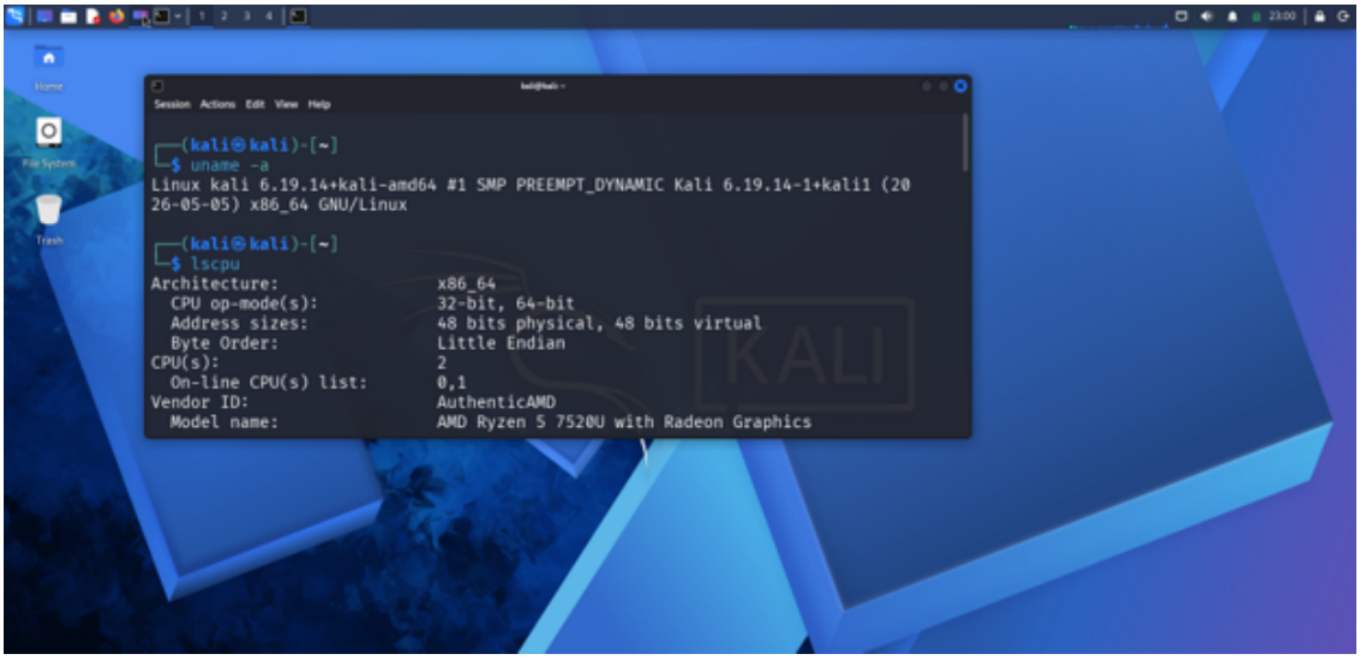


Figure 2. Kernel information and the visible portion of the CPU output.

Linux kernel release: `6.19.14+kali-amd64`.

Architecture: `x86_64`; the CPU supports 32-bit and 64-bit modes.

Processor model: AMD Ryzen 5 7520U with Radeon Graphics.

Logical processors visible to the VM: 2, with online CPUs numbered 0 and 1.

The screenshot shows the top of the CPU report. Physical core count and clock speed are not visible, so they are not inferred here.

Memory & storage

`free -h` reports memory usage; `df -h` reports filesystem space. The `-h` option formats sizes for easier reading.



Figure 3. Memory and disk usage at the time of the exercise.

Total VM RAM: 1.9 GiB (displayed as `1.9Gi`).

Memory available: 1.2 GiB; memory used was 789 MiB.

Main partition: `/dev/sda1`, mounted at `/`.

Available main-partition disk space: 59G, as displayed by `df -h`.

Disk usage: 79G total, 16G used and 21% utilization.

The screenshot also shows 953 MiB of swap, with 0 bytes used. These figures describe the captured moment and may change as the system runs.

Uptime & system activity

`uptime` reports how long the system has been running, the number of logged-in users and recent load averages.

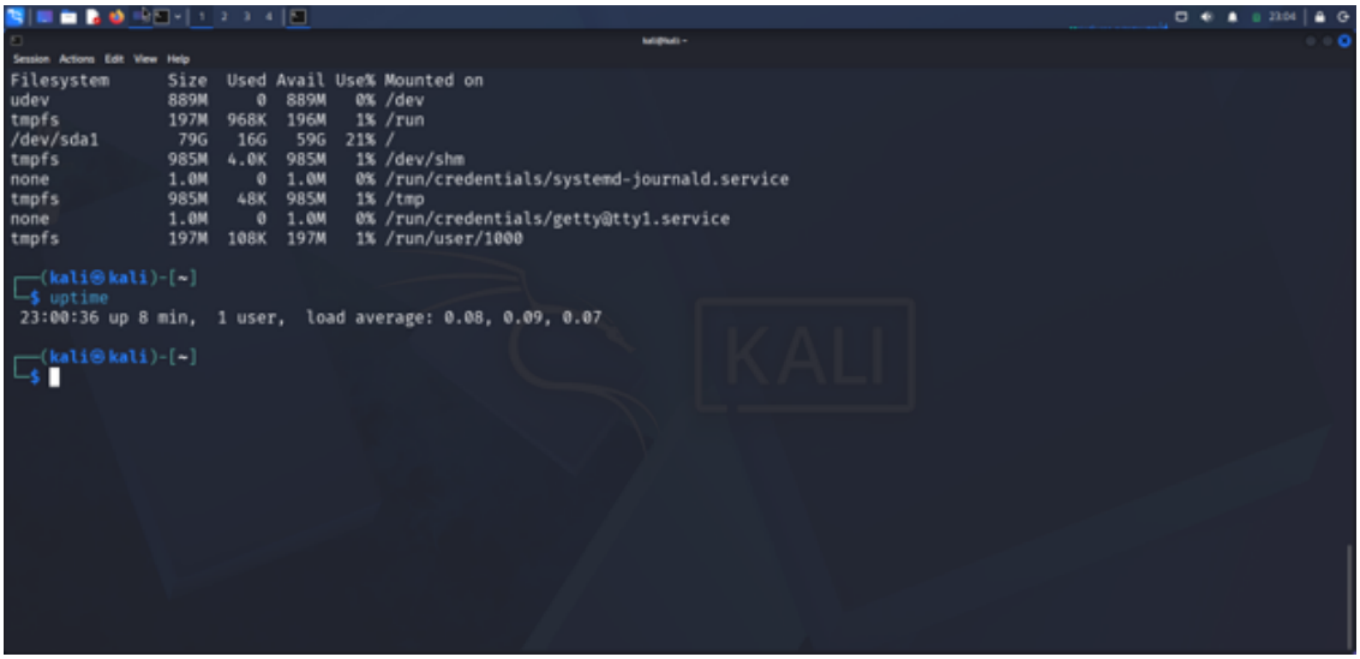


Figure 4. Uptime output following the disk-space report.

System uptime: 8 minutes.

Logged-in users: 1.

Load averages: 0.08 over 1 minute, 0.09 over 5 minutes and 0.07 over 15 minutes.

Load averages describe the average number of tasks running or waiting in an uninterruptible state; they are not CPU-usage percentages. The recorded values indicate light activity during this exercise.

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Network connectivity tests

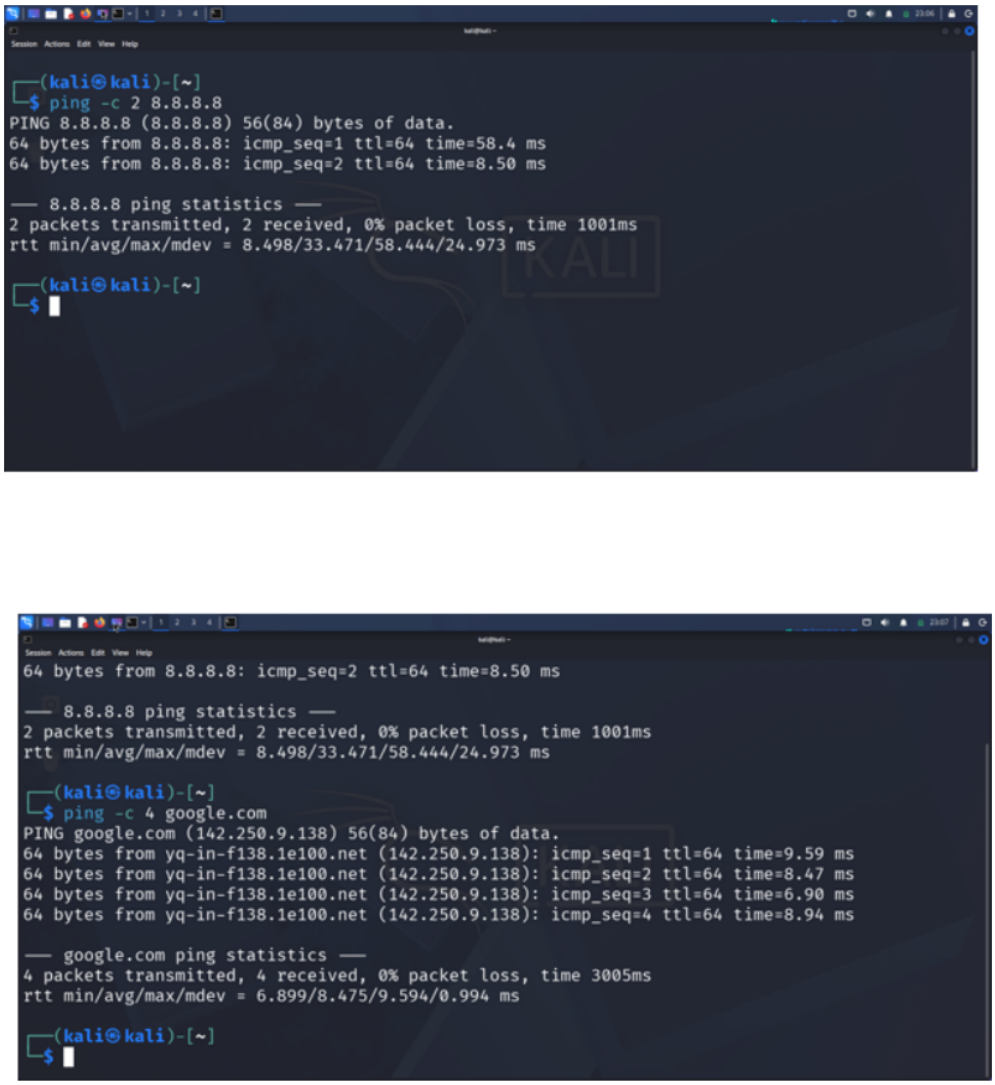


Figure 5. Complete IP-address and domain-name ping tests.

Known IP: `ping -c 2 8.8.8.8` returned 2 of 2 packets with **0% loss**. Average round-trip time: **33.471 ms**.

Domain: `ping -c 4 google.com` resolved to **142.250.9.138** and returned 4 of 4 packets with **0% loss**. Average round-trip time: **8.475 ms**. The `-c 4` option sends four echo requests and then stops.

Comparison: The domain test was faster on average. These are different destinations and may follow different routes. The first IP-test reply took 58.4 ms, raising its average; these small samples do not show that domain names are inherently faster.

Conclusion: The results demonstrate successful connectivity to both tested destinations and successful DNS resolution, alongside practice with Linux identity, hardware, memory, storage and uptime commands.